CERTPATH PUBLISHING · FORMULA CHEAT SHEET

CSP Exam — Quick Reference

Certified Safety Professional · BCSP current blueprint

📊 SAFETY METRICS

Incident Rates

$$TRIR = (\text{recordables} \times 200{,}000) / \text{hrs}$$

$$DART = (\text{DART cases} \times 200{,}000) / \text{hrs}$$

$$Severity = (\text{lost days} \times 200{,}000) / \text{hrs}$$

- 200,000 = 100 emp × 40 hr × 50 wk.

Risk Management

$$Risk = Severity \times Probability$$

- **Hierarchy of Controls** (best → last):
- 1. Elimination 2. Substitution 3. Engineering
- 4. Administrative 5. PPE

🔊 NOISE & VENTILATION

Noise — 1910.95

- **Action level:** 85 dBA TWA → hearing program.
- **PEL:** 90 dBA TWA over 8 hr.
- **Exchange rate:** 5 dB (OSHA); 3 dB (NIOSH).
- Halve exposure time per +5 dB (OSHA).

$$D = 100 \times \Sigma(C_i / T_i)$$

Ventilation

$$Q\ (cfm) = V \times A$$

$$Q_{dilution} = (G \times K) / C_{TLV}$$

- Local exhaust > dilution for high-toxicity.
- Capture velocity: 50–500 fpm (source-dependent).

💪 ERGONOMICS & FIRE

NIOSH Lifting Equation

$$RWL = 51 \times HM \times VM \times DM \times AM \times FM \times CM$$

$$LI = \text{Load Weight} / RWL$$

- LI > 1 → elevated injury risk.
- LI > 3 → most workers at risk.

Fire & Life Safety

- **Classes:** A-solids · B-liquids · C-electrical · D-metals · K-kitchen.
- Travel distance (non-sprinklered): 200 ft typical.
- Exit width: ≥ 32" clear; 0.2"/occupant level, 0.3" stairs.

🧪 INDUSTRIAL HYGIENE

Exposure Limits

- **PEL:** OSHA enforceable.
- **TLV:** ACGIH recommended.
- **STEL:** 15-min max.
- **IDLH:** immediately dangerous.

TWA Sampling

$$TWA = \Sigma(C_i \times T_i) / \Sigma T_i$$

- Personal samples > area samples.
- Calibrate before AND after.

Respirator APF

Type	APF
Half-face APR	10
Full-face APR	50
PAPR (loose)	25

SCBA (P-D)	10,000

📚 **Key regs:** OSHA 1910 (general industry) · 1926 (construction) · 1904 (recordkeeping) · 1200 (HazCom/GHS) · 1910.147 (LOTO) · 1910.146 (Confined Space) · 1910.134 (Respiratory) · 1910.120 (HAZWOPER).